PROJECT REQUIREMENT SPECIFICATION

On-Demand Roadside Assistance Mobile Application Blueprint

Document Version: 1.0	**Target Platforms:** Android, iOS, Web Admin
Core Architecture: Microservices / GPS-Driven	**Primary Monetization:** Commission-Based Split

1. Executive Executive Summary & Concept Overview

The proposed application is a real-time, hyper-local, on-demand marketplace connecting stranded vehicular operators directly to a robust decentralized network of automotive mechanics, specialized towing drivers, and logistical transit runners. Designed for seamless, end-to-end digital orchestration, the solution removes structural friction from vehicle emergencies spanning multi-tier configurations (Two-Wheelers, Four-Wheelers, Heavy Duty Commercial Trucks). By applying location-aware geofencing algorithms, the application minimizes dispatch lag, establishes total transaction transparency, and safeguards platform trust via automated operational checking architectures.

2. System Architecture & Modular Platform Access

The platform is structurally segregated into four dedicated architectural access modes designed to align with user operational workflows:

- **Customer Application Interface:** Provides multi-vehicle baseline profiling (Model, Brand, License Plate). Features an automated background GPS telemetry listener to fetch exact live coordinates. Houses the primary upfront billing presentation layout and real-time operational map progression screens.
- **Mechanic Operational Terminal:** Incorporates a critical, battery-optimized background location tracking toggle allowing field technicians to change availability status ('Go Online/Offline'). Includes integrated navigation maps routing directly to breakdown locations and interactive digital job-card estimators.
- **Towing & Crane Logistics Module:** A highly custom functional variant of the driver layout mapped to accept variable structural payloads based on physical truck categorization. Facilitates multi-point route tracking.
- **Admin Command & Control Dashboard:** Built specifically for the platform operator (You). Provides live operational maps displaying active geo-requests, pending system onboarding validation states for newly registered field agents, and financial ledgers mapping net revenue generation and platform commission splits.

3. Core Services Strategy & Base Fleet Matrix

The platform organizes services across three clear tiers of heavy, medium, and light consumer vehicles to optimize dispatch routing rules and manage base service expectations:

Core Services Available	■■ Two-Wheeler (Bike)	■ Four-Wheeler (Car)	■ Heavy Vehicle (Truck)
Puncture Assistance	Fixed Base Fee + Distance Charge	Fixed Base Fee + Distance Charge	Fixed Base Heavy Fee + Distance
Mechanical Breakdown	■250 Base Diagnostic + Distance	■500 Base Diagnostic + Distance	■1,000 Base Diagnostic + Distance
Towing / Crane Fleet	Flatbed / Standard Chain Lift	Integrated Wheel-Lift / Flatbed	Heavy Utility Boom Crane Winch
Fuel Delivery Supply	Emergency Premium Petrol	Petrol / Multi-Grade Diesel	High-Volume Bulk Diesel Supply

4. Algorithmic Pricing Engine & Distance Surcharges

Total Client Invoice Formula:

Invoice = [Base Service / Diagnostic Fee] + (Total Distance (km) × ■15)

Operational Use-Cases:

- **Scenario A (Bike Puncture at 10km):** If the flat puncture repair base configuration is set to ■150, the distance computation applies: 10 km × ■15 = ■150. Combined invoice output reads ■**300**.
- **Scenario B (Car Mechanical Diagnostics at 20km):** Initial setup utilizes the fixed baseline diagnostic checking requirement of ■500. The logistics calculation computes: 20 km × ■15 = ■300. Combined invoice output reads ■**800**.

5. Monetization Framework & Revenue Generation Model

- **Primary Marketplace Commissions:** A fixed percentage split (suggested baseline standard: 15% to 20%) is calculated automatically from the total baseline diagnostic or mechanical repair invoice fee. Distance

traveling payouts are typically pushed entirely to the service providers to cover immediate fuel overhead.

- **Fuel Convenience Margins:** Emergency fuel delivery features a dual revenue model combining flat platform container delivery processing charges with premium per-litre fuel surcharges added over current local market daily standard indexes.
- **B2B Premium Mechanic SaaS Subscriptions:** Registered mechanics or commercial towing fleets can optionally opt into a monthly platform SaaS tier. Subscribers gain access to 'Priority Dispatch Alerts' providing a 30-second structural head-start to claim jobs in hyper-congested zones before broadcast to non-subscribers.
- **Hyper-Targeted In-App Advertisement Real-Estate:** Structural delivery screens and real-time mechanics tracking pages house specific programmatic layout positions reserved for automotive insurance renewals, spare parts providers, battery retailers, and major tire manufacturing ecosystems.